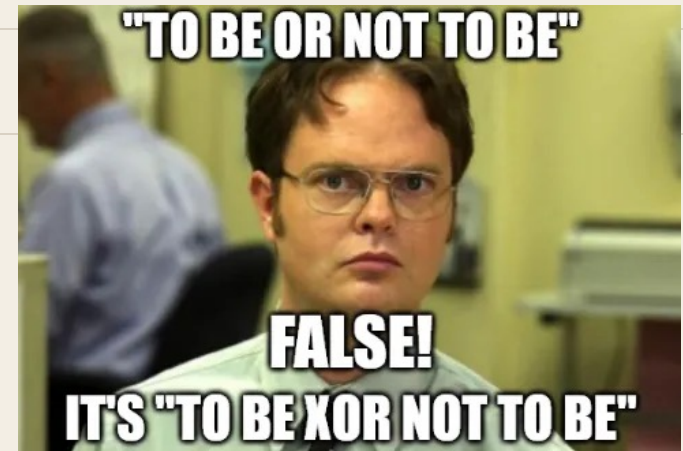

Propositional Logic

Six operators, four-row tables, and the laws that let you skip the tables.

Today

AGENDA

- 01 What a proposition is
- 02 Negation, conjunction, disjunction, exclusive or
- 03 Conditionals, and the three statements people confuse with them
- 04 Biconditionals
- 05 The equivalence laws



Propositions

BUILDING BLOCKS

DEFINITION

A **proposition** is a declarative sentence — a sentence that declares a fact — that is either true or false, but not both.

We write propositions as letters: p , q , r .

Each carries exactly one bit of information: its **truth value**, T or F.

Which of These Are Propositions?

BUILDING BLOCKS

Columbus is the capital of Ohio.

Proposition – true

$1 + 1 = 3$.

Proposition – false

What time is the midterm?

Not a proposition – a question

Please read the syllabus.

Not a proposition – a command

$x + 1 = 2$.

Not yet – depends on x

That last one gets a name next lecture: a **predicate**.

Negation

OPERATORS

DEFINITION

Let p be a proposition. The **negation** of p , written $\neg p$, is the statement “It is not the case that p .” Its truth value is always the opposite of p .

Read aloud as *not p*. Also written p^- .

p	$\neg p$
T	F
F	T

Conjunction

OPERATORS

DEFINITION

The **conjunction** of p and q , written $p \wedge q$, is the proposition “ p and q .”
It is true when both are true, and false otherwise.

One false input poisons the whole thing.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Disjunction

OPERATORS

DEFINITION

The **disjunction** of p and q , written $p \vee q$, is the proposition “ p or q .” It is false only when both are false.

This is *inclusive* or: both counts as yes.

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Exclusive Or

OPERATORS

DEFINITION

The **exclusive or** of p and q , written $p \oplus q$, is true when exactly one of the two is true, and false otherwise.

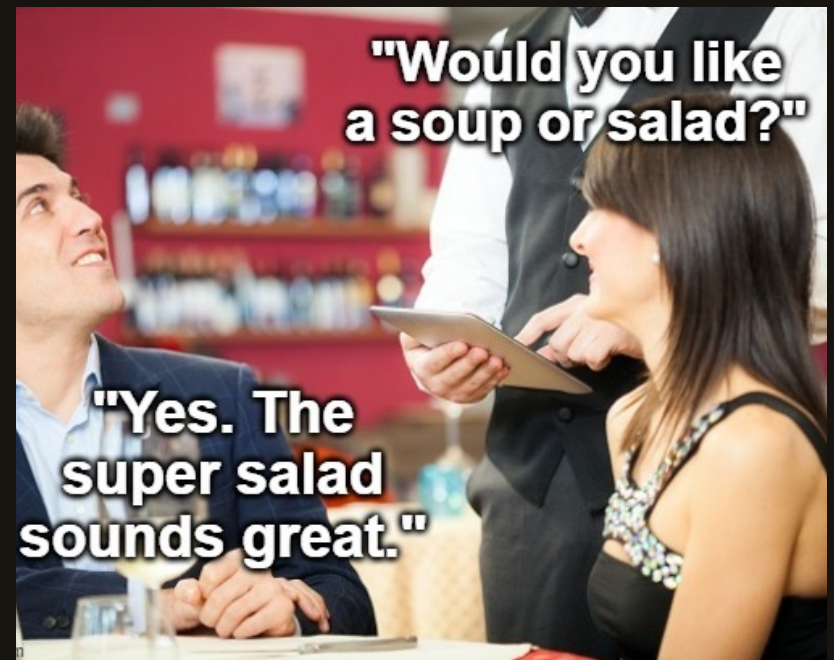
Only the first row differs from $\vee$. That one row causes most of the confusion in this course.

p	q	$p \oplus q$
T	T	F
T	F	T
F	T	T
F	F	F

OR, TWO WAYS

“Soup or salad?” “Yes.”

The waiter meant $\oplus$. You answered $\vee$. Both of you are right, which is exactly why mathematics does not let you say “or” without picking a symbol.



Operators in Code

OPERATORS

Logic	Java / C / C++	Python
$\neg p$	<code>!p</code>	<code>not p</code>
$p \wedge q$	<code>p && q</code>	<code>p and q</code>
$p \vee q$	<code>p q</code>	<code>p or q</code>
$p \oplus q$	<code>p ^ q</code>	<code>p != q</code>
$p \leftrightarrow q$	<code>p == q</code>	<code>p == q</code>
$p \rightarrow q$	<code>!p q</code>	<code>(not p) or q</code>

No mainstream language ships an implication operator.

You write `!p || q` instead — which is a law we will prove in about twenty minutes.

Conditional Statements

CONDITIONALS

DEFINITION

The **conditional statement** $p \rightarrow q$ is the proposition “if p , then q .” It is false when p is true and q is false, and true otherwise.

p is the **hypothesis**, q is the **conclusion**.

p	q	$p \rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Vacuous ~~Truth~~

CONDITIONALS

$$F \rightarrow q \equiv T$$

A conditional promises something only about the case where the hypothesis holds. If the hypothesis never holds, the promise was never tested — so it was never broken. Both of these are **true** statements:

01 If I start at quarterback for the Buckeyes, then I can bench 500 pounds.

02 If pigs fly, then $1 = 2$.

Nobody is checking, because nobody ever gets to the check.

Twelve Ways to Say $p \rightarrow q$

CONDITIONALS

if p , then q

if p , q

p implies q

p is sufficient for q

a sufficient condition for q is p

p only if q

q when p

q if p

q whenever p

q follows from p

q is necessary for p

q unless $\neg p$

The two in scarlet are the ones that show up on exams.

Example: Maria and Discrete Math

CONDITIONALS

GIVEN

p : Maria learns discrete mathematics.

q : Maria will find a good job.

Express $p \rightarrow q$ in English.

“If Maria learns discrete mathematics, then she will find a good job.”

“Maria will find a good job when she learns discrete mathematics.”

“For Maria to get a good job, it is sufficient for her to learn discrete mathematics.”

“Maria will find a good job unless she does not learn discrete mathematics.”

Converse, Contrapositive, Inverse

CONDITIONALS

ORIGINAL

$$p \rightarrow q$$

The statement you started with.

CONVERSE

$$q \rightarrow p$$

Swap the two sides.

CONTRAPOSITIVE

$$\neg q \rightarrow \neg p$$

Swap *and* negate.

INVERSE

$$\neg p \rightarrow \neg q$$

Negate both sides.

Example: Rain and the Home Team

CONDITIONALS

STEP 1 — REWRITE

“The home team wins whenever it is raining” → “**If it is raining, then the home team wins.**”

CONTRAPOSITIVE

“If the home team does not win, then it is not raining.”

CONVERSE

“If the home team wins, then it is raining.”

INVERSE

“If it is not raining, then the home team does not win.”

The converse claims rain is the *only* way they win. The original claims nothing of the kind.

Only the Contrapositive Is Equivalent

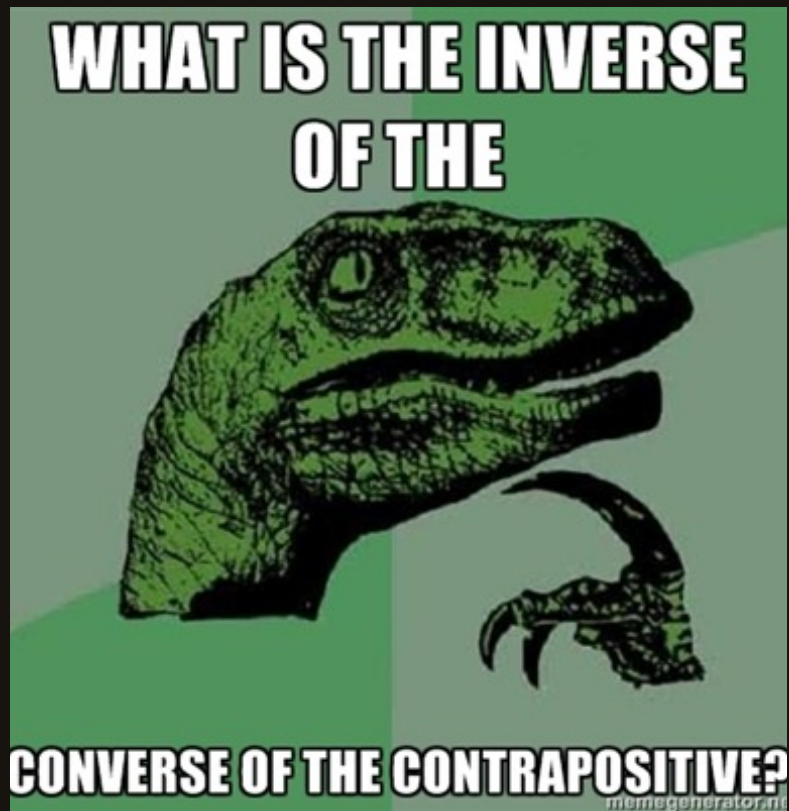
CONDITIONALS

p	q	$p \rightarrow q$	$q \rightarrow p$	$\neg q \rightarrow \neg p$	$\neg p \rightarrow \neg q$
T	T	T	T	T	T
T	F	F	T	F	T
F	T	T	F	T	F
F	F	T	T	T	T

$$p \rightarrow q \equiv \neg q \rightarrow \neg p$$

The converse and the inverse agree with each other — and with the original only by accident.

Every proof by contraposition in Lecture 3 rests on this one column match.



THE CLASSIC ERROR

“All my professors are tired. So all tired people are professors.”

Flipping an implication is not a proof technique. It is a different claim, and usually a false one.

Biconditional Statements

BICONDITIONALS

DEFINITION

The **biconditional** $p \leftrightarrow q$ is the proposition “ p if and only if q .” It is true when p and q have the same truth value, and false otherwise.

It is the exact opposite of $\oplus$. Compare the two tables.

p	q	$p \leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Ways to Say $p \leftrightarrow q$

BICONDITIONALS

01 “ p is necessary and sufficient for q ”

02 “if p then q , and conversely”

03 “ p iff
 q ”

“iff” is not a typo. Nobody in this field has ever fixed it.

Precedence and Parentheses

NOTATION

$\neg$ then $\wedge$ then $\vee$ then $\rightarrow$ then $\leftrightarrow$

READS AS

$\neg p \wedge q \rightarrow r \vee s$

means $((\neg p) \wedge q) \rightarrow (r \vee s)$

Knowing this is worth two points. Writing the parentheses anyway is worth the whole problem.

Equivalence Laws I

THE LAWS

IDENTITY

$$p \wedge T \equiv p$$

$$p \vee F \equiv p$$

DOMINATION

$$p \vee T \equiv T$$

$$p \wedge F \equiv F$$

IDEMPOTENT

$$p \vee p \equiv p$$

$$p \wedge p \equiv p$$

COMMUTATIVE

$$p \vee q \equiv q \vee p$$

$$p \wedge q \equiv q \wedge p$$

NEGATION

$$p \vee \neg p \equiv T$$

$$p \wedge \neg p \equiv F$$

DOUBLE NEGATION

$$\neg(\neg p) \equiv p$$

Two nots cancel.

Equivalence Laws II

THE LAWS

DISTRIBUTIVE $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$ $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

ABSORPTION $p \vee (p \wedge q) \equiv p$ $p \wedge (p \vee q) \equiv p$

ASSOCIATIVE $p \vee (q \vee r) \equiv (p \vee q) \vee r$ $p \wedge (q \wedge r) \equiv (p \wedge q) \wedge r$

De Morgan's Laws

THE LAWS

$$\neg (p \wedge q) \equiv \neg p \vee \neg q$$

$$\neg (p \vee q) \equiv \neg p \wedge \neg q$$

Push the negation inside, and flip the connective on the way in.

IN CODE

```
// hard to read
if (!(user != null && user.isActive))

// same thing, De Morgan'd
if (user == null || !user.isActive)
```


The Implication Laws

THE LAWS

$p \rightarrow q \equiv \neg p \vee q$ Turns an arrow into an $\vee$, so the rest of the laws can reach it.

$\neg (p \rightarrow q) \equiv p \wedge \neg q$ The negation of an implication is *not* an implication. It is an $\wedge$.

Some texts call the first one material implication. Whatever you call it, you will use it more than any other law this semester.

More Equivalences

THE LAWS

$$27 \quad (p \leftrightarrow q) \equiv (p \rightarrow q) \wedge (q \rightarrow p)$$

$$28 \quad p \rightarrow (q \wedge r) \equiv (p \rightarrow q) \wedge (p \rightarrow r)$$

$$29 \quad p \rightarrow (q \vee r) \equiv (p \rightarrow q) \vee (p \rightarrow r)$$

$$30 \quad (p \wedge q) \rightarrow r \equiv (p \rightarrow r) \vee (q \rightarrow r)$$

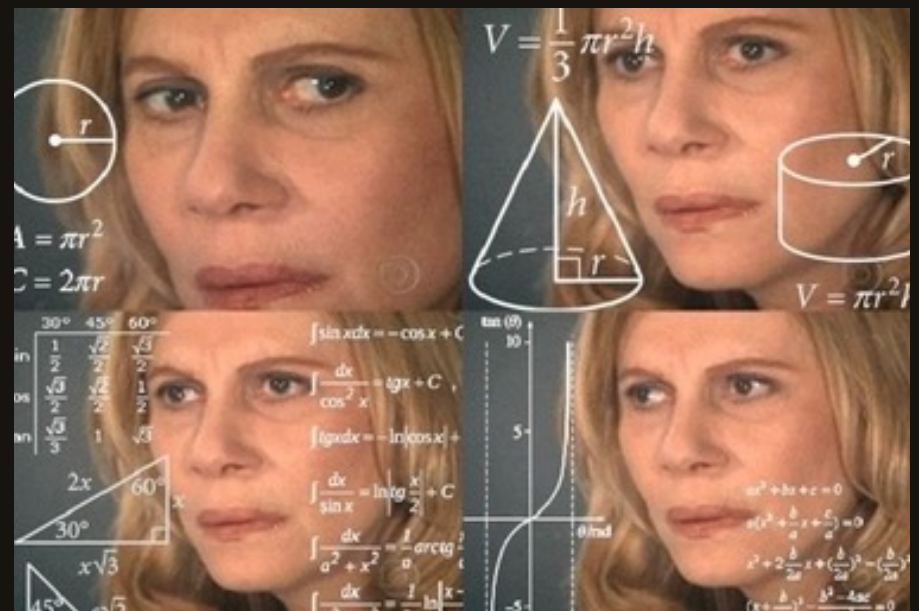
$$31 \quad (p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$$

Every one of these is the implication law plus distribution. None of them needs to be memorized — but 30 and 31 are worth staring at, because the connective flips.

THIRTY-ONE LAWS LATER

You do not have to memorize these.

You have to recognize them. Three of them — implication, De Morgan, distribution — will carry ninety percent of the work.



Common Mistakes

BEFORE YOU GO

- 01 Reading $p \rightarrow q$ as “ p causes q .” Implication says nothing about cause.
- 02 Using the converse where the contrapositive belongs. Only $\neg q \rightarrow \neg p$ is equivalent.
- 03 Reading $\vee$ as exclusive. In this course “or” always includes both, unless you see $\oplus$.
- 04 Negating $p \rightarrow q$ as $\neg p \rightarrow \neg q$. The negation is $p \wedge \neg q$, and it is not a conditional at all.

Extra Credit

BEFORE YOU GO

Let p, q, r be propositions. Which of these hold? Prove it, or produce one row that breaks it.

$$p \oplus (q \vee r) \equiv (p \oplus q) \vee (p \oplus r)$$

$$p \oplus (q \wedge r) \equiv (p \wedge q) \vee (p \wedge r)$$

$$p \oplus (q \oplus r) \equiv (p \oplus q) \oplus r$$

One row is enough to kill an equivalence. Finding it is faster than writing all eight.

Recap

LECTURE 01

- 01 A proposition is a declarative sentence with exactly one truth value.
- 02 Six operators, six truth tables. $\vee$ is inclusive; $\oplus$ is the exclusive one.
- 03 $p \rightarrow q$ is false in exactly one row: true hypothesis, false conclusion.
- 04 Of the converse, the contrapositive, and the inverse, only the contrapositive is equivalent.
- 05 The laws let you rewrite instead of enumerate. $p \rightarrow q \equiv \neg p \vee q$ is the one to keep in your head.

Next: tautologies, equivalence proofs without tables, and quantifiers.